

Double-click (or enter) to edit

```
# 1.Acquire
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import ipywidgets as widgets
from IPython.display import display

sns.set(style="whitegrid")

# Upload heart.csv
from google.colab import files
uploaded = files.upload()

# Load dataset
df = pd.read_csv("heart.csv")
df.head()
```



No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please

rerun this cell to enable.

Saving heart.csv to heart (1).csv

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Sl
0	40	M	ATA	140	289	0	Normal	172	N	0.0	
1	49	F	NAP	160	180	0	Normal	156	N	1.0	
2	37	M	ATA	130	283	0	ST	98	N	0.0	
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	
4	54	M	NAP	150	195	0	Normal	122	N	0.0	

```
# 2. Parse
```

```
# Check data types and nulls
df.info()
df.describe()
```



What can I help you build?



```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Age                   918 non-null    int64
 1   Sex                   918 non-null    object
 2   ChestPainType         918 non-null    object
 3   RestingBP             918 non-null    int64
 4   Cholesterol           918 non-null    int64
 5   FastingBS            918 non-null    int64
 6   RestingECG           918 non-null    object
 7   MaxHR                918 non-null    int64
 8   ExerciseAngina        918 non-null    object
 9   Oldpeak              918 non-null    float64
10   ST_Slope              918 non-null    object
11   HeartDisease          918 non-null    int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB

```

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisease
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	0.233115	136.809368	0.887364	0.553377
std	9.432617	18.514154	109.384145	0.423046	25.460334	1.066570	0.497414
min	28.000000	0.000000	0.000000	0.000000	60.000000	-2.600000	0.000000
25%	47.000000	120.000000	173.250000	0.000000	120.000000	0.000000	0.000000
50%	54.000000	130.000000	223.000000	0.000000	138.000000	0.600000	1.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.500000	1.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000	6.200000	1.000000

```
# 3.Filter
```

```
# Select key features
```

```
filtered_df = df[['HeartDisease', 'Age', 'Sex', 'ChestPainType', 'Cholesterol', 'MaxHR']].dropna()
```

```
# Convert categorical features to numeric
```

```
filtered_df['Sex'] = filtered_df['Sex'].map({'M': 1, 'F': 0})
```

```
filtered_df['ChestPainType'] = filtered_df['ChestPainType'].astype('category').cat.codes
```

```
filtered_df.head()
```

	HeartDisease	Age	Sex	ChestPainType	Cholesterol	MaxHR
0	0	40	1	1	289	172
1	1	49	0	2	180	156
2	0	37	1	1	283	98
3	1	48	0	0	214	108
4	0	54	1	2	195	122

```
# 4 Mine
```

```
# Count of HeartDisease
```

```
print("Heart Disease Presence:\n", filtered_df['HeartDisease'].value_counts())
```

```
# Average cholesterol by Heart Disease
```

```
print("\nAverage Cholesterol by HeartDisease:\n", filtered_df.groupby('HeartDisease')['Cholesterol'].mean())
```

```
# Correlation matrix
```

```
filtered_df.corr(numeric_only=True)
```

```

Heart Disease Presence:
HeartDisease
1      508
0      410
Name: count, dtype: int64

```

```

Average Cholesterol by HeartDisease:
HeartDisease
0      227.121951
1      175.940945
Name: Cholesterol, dtype: float64

```

	HeartDisease	Age	Sex	ChestPainType	Cholesterol	MaxHR
HeartDisease	1.000000	0.282039	0.305445	-0.386828	-0.232741	-0.400421
Age	0.282039	1.000000	0.055750	-0.077150	-0.095282	-0.382045
Sex	0.305445	0.055750	1.000000	-0.126559	-0.200092	-0.189186
ChestPainType	-0.386828	-0.077150	-0.126559	1.000000	0.067880	0.289123
Cholesterol	-0.232741	-0.095282	-0.200092	0.067880	1.000000	0.235792
MaxHR	-0.400421	-0.382045	-0.189186	0.289123	0.235792	1.000000

```
# 5 Represent
```

```

# Bar chart
plt.figure(figsize=(6, 4))
sns.countplot(data=filtered_df, x='ChestPainType', hue='HeartDisease')
plt.title('Heart Disease by Chest Pain Type')
plt.xlabel('Chest Pain Type')
plt.ylabel('Count')
plt.show()

```

```

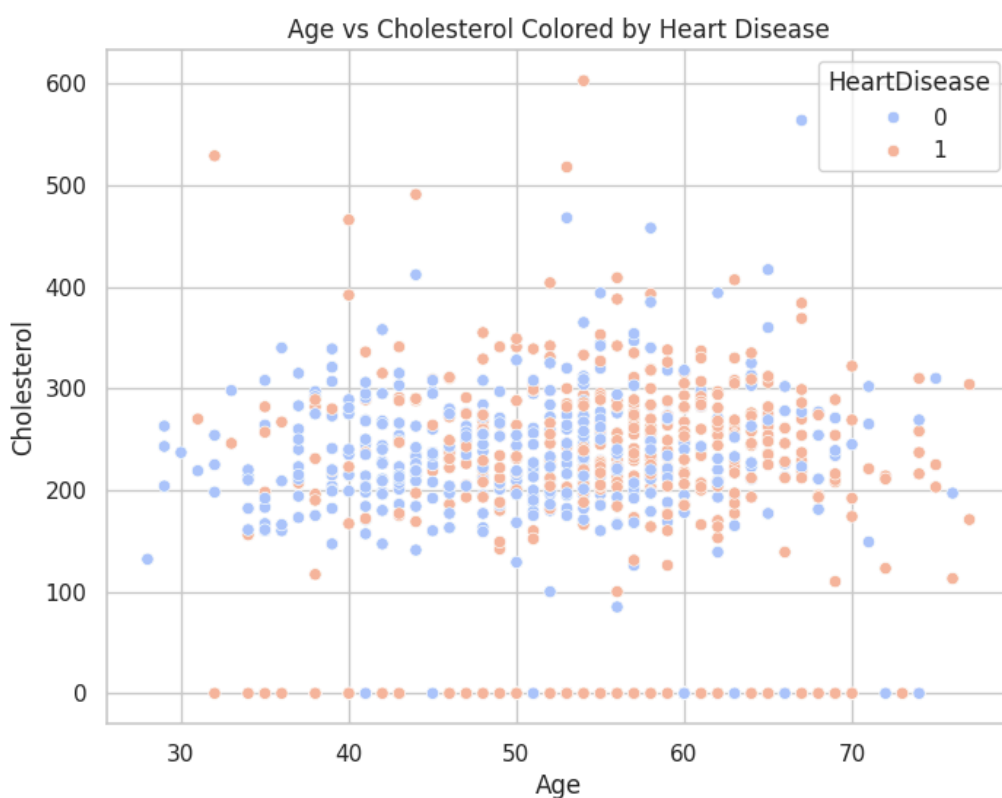
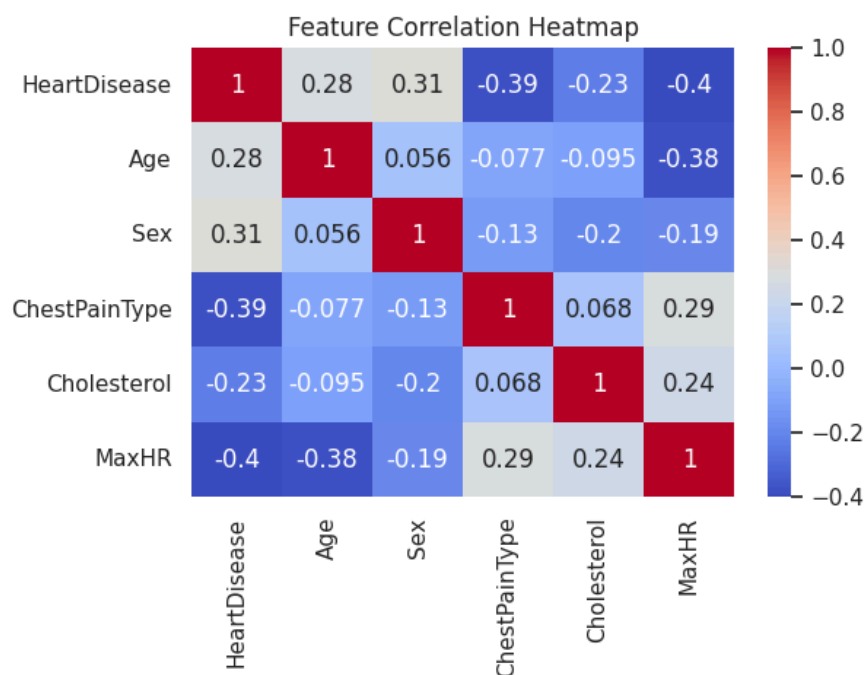
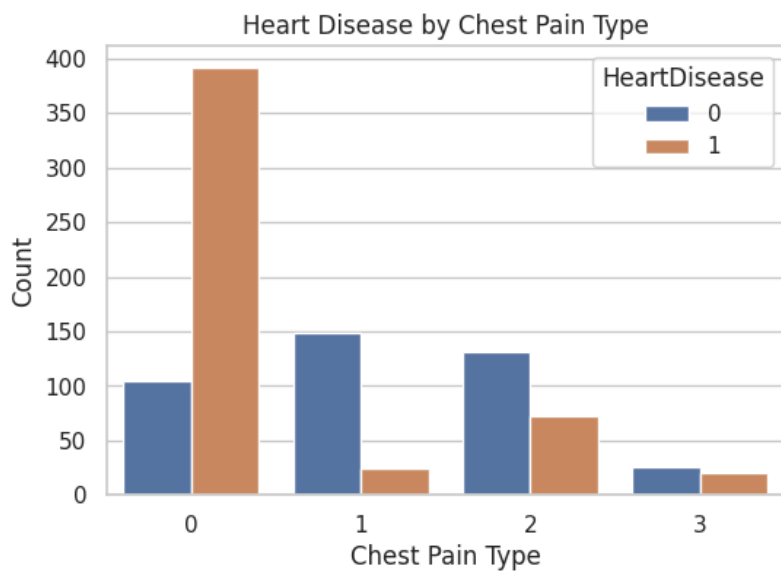
# Heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(filtered_df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Feature Correlation Heatmap')
plt.show()

```

```

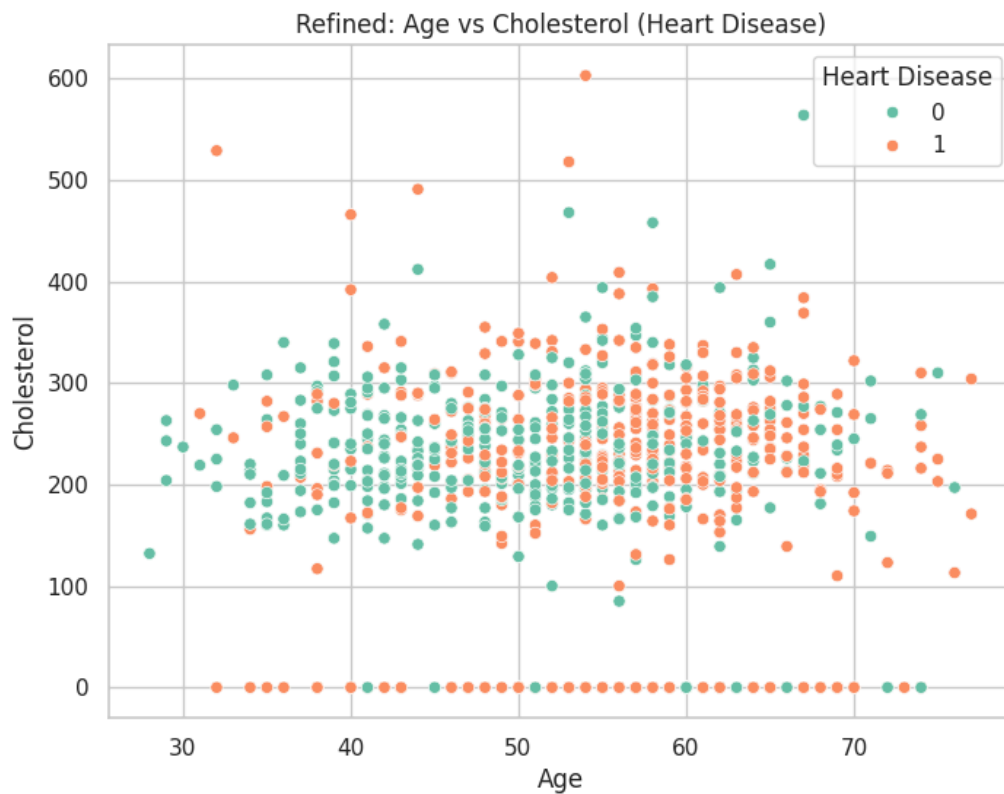
# Scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(data=filtered_df, x='Age', y='Cholesterol', hue='HeartDisease', palette='coolwarm')
plt.title('Age vs Cholesterol Colored by Heart Disease')
plt.xlabel('Age')
plt.ylabel('Cholesterol')
plt.show()

```



6 Refine

```
plt.figure(figsize=(8, 6))
sns.scatterplot(data=filtered_df, x='Age', y='Cholesterol', hue='HeartDisease', palette='Set2')
plt.title('Refined: Age vs Cholesterol (Heart Disease)')
plt.xlabel('Age')
plt.ylabel('Cholesterol')
plt.grid(True)
plt.legend(title='Heart Disease')
plt.show()
```



7 Interact

```
import pandas as pd
import ipywidgets as widgets
from IPython.display import display

# Load the data (adjust the path if needed)
df = pd.read_csv('heart.csv')

# If you want to filter the DataFrame, do it here.
# For now, let's use the full DataFrame as filtered_df:
filtered_df = df

# Create the dropdown widget
heart_disease_dropdown = widgets.Dropdown(
    options=filtered_df["HeartDisease"].unique(),
    description="Heart Disease:"
)

# Define the function to display data based on dropdown selection
def show_heart_data(status):
    display(filtered_df[filtered_df["HeartDisease"] == status].head())

# Create the interactive widget
widgets.interact(show_heart_data, status=heart_disease_dropdown)
```



Heart Dise... 0

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0
5	39	M	NAP	120	339	0	Normal	170	N	0.0	Up	0
6	45	F	ATA	130	237	0	Normal	170	N	0.0	Up	0

```
show_heart_data
def show_heart_data(status)
```